

Saurav Kumar Choudhary

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EDUCATION

Institute of Bioinformatics University of Georgia Athens, Georgia, USA	Aug 2022 – Present
Doctor of Philosophy (Ph.D. in Bioinformatics)	GPA: 4.0/4.0
Indraprastha Institute of Information Technology Delhi, India	Jul 2020 – Jun 2022
Master of Technology in Computational Biology	GPA: 8.25/10
Amity Institute of Biotechnology Noida, India	Jul 2015 – Jun 2019
Bachelor of Technology in Biotechnology	GPA: 7.02/10

RESEARCH EXPERIENCE

Institute of Bioinformatics University of Georgia Athens, Georgia, USA	Aug 2022 – Present
Graduate Student Researcher (Bioinformatics)	
Advisor – Dr. Kaixiong Ye	
- Integrated human atlas-scale single-cell transcriptomics and epigenomic datasets with GWAS summary statistics to prioritize trait-relevant cell types underlying circulatory fatty acid metabolism. - Applying Deep learning based framework to infer gene regulatory networks from single-cell multiome (scRNA-seq and scATAC-seq) data using atlas-scale human datasets. - Built an end-to-end de novo genome assembly and annotation pipeline for <i>Hymenolepis diminuta</i> using PacBio HiFi reads, RepeatModeler/RepeatMasker, and BRAKER3, including synteny analysis, producing a chromosome-scale reference genome; available as a bioRxiv preprint, with the assembly publicly available (GCA_056153285.1) - Developing a scRNA-seq atlas of <i>Hymenolepis diminuta</i> by clustering, annotating, and integrating cell populations across multiple samples. - Analyzed large-scale scRNA-seq datasets from collaborators, performing QC, alignment, quantification, normalization, clustering, and differential expression analysis.	
Indraprastha Institute of information Technology Delhi, India	Jun 2021- Jun 2022
Student Researcher	
Advisor – Dr. Vibhor Kumar	
- Developed pathway based feature representations from scRNA-seq data to improve interpretability and reduce feature dimensionality for machine-learning models. - Applied machine learning models to pathway-level scRNA-seq features to identify conserved signatures across stem cell types and enable robust cell-state prediction under batch effects.	
Eminent Biosciences Indore, India	Feb 2020 – Jul 2020
Bioinformatics Analyst	
Advisor - Dr. Anuraj Nayarisseri	
- Performed bulk RNA-seq differential expression analysis of MCF-7 breast cancer and normal samples using R-based workflows. - Built and executed reproducible NGS analysis scripts in Python, R, and Bash on Linux systems.	
Amity Institute of Biotechnology Noida, India	Jan 2019 – Apr 2019
Student Researcher (Structural Bioinformatics)	
Advisor – Dr. Archana Chaturvedi	
Performed protein modeling, active site prediction, molecular docking, model visualization and ADMET property analysis for GCC-ECD receptor.	
CSIR- Institute of Genomics and Integrative Biology New Delhi, India	Jan 2019 – Apr 2019
Research Intern	
Advisor – Dr. S Ramachandarn	
- Conducted text-mining of biomedical literature using R (pubmed.mineR) to curate candidate genes associated with Type 2 Diabetes. - Identified and reported 30 disease-associated genes with literature-based supporting evidence.	

PUBLICATIONS

- **Choudhary SK**, Sundaresha N, Ye K, Bergman CM, Rozario T. Assembly of a high-quality reference genome for the rat tapeworm *Hymenolepis diminuta*. bioRxiv [Preprint]. 2026 Jun 23:2026.06.23.733678. doi: 10.64898/2026.06.23.734100.
- Dillon, K. C., Sprong, H., Ronai, I., **Choudhary, S.**, Grant, C., Jr., de Paula Baptista, R., Ray, D. A., & Glenn, T. C. (2026). Optimizing genome assembly, chromosome synteny, and genetic variant discovery from Oxford Nanopore sequences of *Dermacentor reticulatus* ticks. bioRxiv. <https://doi.org/10.64898/2026.09.18.752137>.
- Fu Z, Huang Z, Xu H, Liu Q, Li J, Song K, Deng Y, Tao Y, Zhang H, Wang P, Li H, Sheng Y, Zhou A, Han L, Fu Y, Wang C, **Choudhary SK**, Ye K, Veggiani G, Li Z, August A, Huang W, Shan Q, Peng H. IL-2-inducible T cell kinase deficiency sustains chimeric antigen receptor T cell therapy against tumor cells. J Clin Invest. 2024 Nov 26;135(4):e178558. doi: 10.1172/JCI178558. PMID: 39589809; PMCID: PMC11827851.
- Yu W, Kastri ME, Ishan M, **Choudhary SK**, Rashid MM, Kramer N, Do HGT, Wang Z, Xu T, Schwabe RF, Ye K, Adameyko I, Liu HX. The duct of von Ebner's glands is a source of *Sox10*⁺ taste bud progenitors and susceptible to pathogen infections. Front Cell Dev Biol. 2024 Aug 23;12:1460669. doi: 10.3389/fcell.2024.1460669. PMID: 39247625; PMCID: PMC11377339.
- Gupta N †, **Choudhary SK †**, Bhagat N, Karthikeyan M, Chaturvedi A. In Silico Prediction, Molecular Docking and Dynamics Studies of Steroidal Alkaloids of *Holarrhena pubescens* Wall. ex G. Don to Guanylyl Cyclase C: Implications in Designing of Novel Antidiarrheal Therapeutic Strategies. Molecules. 2021 Jul 8;26(14):4147. doi: 10.3390/molecules26144147. PMID: 34299422; PMCID: PMC8305770.
- Qureshi S, Khandelwal R, Madhavi M, Khurana N, Gupta N, **Choudhary SK**, Suresh RA, Hazarika L, Srija CD, Sharma K, Hindala MR, Hussain T, Nayarisseri A, Singh SK. A Multi-target Drug Designing for BTK, MMP9, Proteasome and TAK1 for the Clinical Treatment of Mantle Cell Lymphoma. Curr Top Med Chem. 2021;21(9):790-818. doi: 10.2174/1568026621666210119112336. PMID: 33463471.

SKILLS

Genomics & Transcriptomics: Single-cell RNA-seq, Single-cell ATAC-seq, Spatial transcriptomics, bulk RNA-seq, Genome Analysis: Genome assembly and annotation, GWAS integration.

Programming Languages: R, Python

Scripting: Bash (UNIX shell)

R & Bioconductor: Seurat, DESeq2, ArchR, Tidyverse, dplyr, ggplot2, stringr

Python Libraries: Scanpy, NumPy, Pandas, Matplotlib, PyTorch, scikit-learn

Workflow & Reproducibility: Git version control, documented and reproducible analysis pipelines

High-Performance Computing: Linux-based HPC environments, SLURM workload manager, job scheduling and resource management

Bioinformatics Tools: Cell Ranger, STAR, HISAT2 (scRNA/RNA-seq); minimap2, Hifiasm, BRAKER3 (long-read assembly and annotation); GENESPACE (comparative genomics)

HONORS & PROFESSIONAL DEVELOPMENT

- 2nd Place, Poster Presentation Competition - Institute of Bioinformatics (IOB) Symposium, University of Georgia, September 2026
- Peer Reviewer, PLOS Pathogens.
- Selected Participant, Duke-Chen Institute Joint Boot Camp on AI & AI-Accelerated Medical Research, Duke University (May 2025).
- Awarded "Best Minimalistic Poster Design," Institute of Bioinformatics (IOB) Spring Retreat, April 2025.
- Genetics Retreat Poster Award, September 2025.
- Selected Participant, UCLA Computational Genomics Summer Institute (CGSI), 2024.
- Secretary, Indian Student Association, University of Georgia (2023–2024); secured funding and coordinated community events for the UGA campus.
- Selected Participant (Top 25 nationally), B4: Young Scientist Development Course – Big Data in Life Sciences and Healthcare (Harvard University & IBAB, India).
- GATE Fellowship Recipient (All India Rank 1117) — awarded for the full duration of the M.Tech program.
- Poster presentations at the American Society of Human Genetics (ASHG), 2023–2025.
- Completed, R Programming (Coursera – Johns Hopkins University).